



MTP Department
SIMHADRI



Ref:SMTPP/MAINT. PLNG./2022-23/MTP-LMI/508565

Date:11/10/2022(17:23:18)

Subject:LMI ON CERTIFICATION OF CP, AP, SAP FOR PTW SYSTEM

Sub: Approval of LMI on "Certification of SAPs, APs & CPs for PTW System".

LMI on "Certification of SAPs, APs & CPs for PTW System" is reviewed based on LMI/ ISO-00-OGN/GEN/004, REV NO. 1, DATE: OCT2020 & COS-ISO-00-OGN/GEN/004, Rev. No.: 01, Sep 2020 and **no changes are proposed.**

Submitted for approval.

Encl: LMI for approval & OGN Dtd Sep 2022.

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Review of LMI revision done.

Revised LMI (no changes required/incorporated) may please be accorded approval.

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LOCATION MANAGEMENT INSTRUCTION

LMI/ ISO-00-OGN/GEN/004

REVISION NO. 2, DATE: OCT2022

**Certification of SAPs, APs & CPs
for PTW System**

Approved for Implementation: -----

HOP, SIMHADRI

LMI APPROVAL DOCUMENTS

1.0 TITLE OF LMI	LMI on Certification of SAPs, APs & CPs for PTW System
2.a) DOCUMENT CODE b) REVISION/ISSUE NO. / DATE	LMI/ OGN/GEN/004 REV- 1, ISSUE-2, OCT 2022
3.ORIGINAL/REF.DOCUMENT	COS-ISO-00-OGN/GEN/004 Rev: 1, SEPT 2020
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Certification of SAPs, APs & CPs for PTW System

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1.0 INTRODUCTION

In order to strengthen the PTW system, SAP/AP/CP are to be nominated for SIMHADRI station. This LMI offers guidance on implementation of the process

2.0 SUPERSEDED DOCUMENTS

LMI/ ISO-00-OGN/GEN/004, REV-1, OCT 2020

3.0 REFERENCE DOCUMENT

COS-ISO-00-OGN/GEN/004, Rev-1, Sept2020

4.0 OBJECTIVE: To have a System / Process for nomination of Senior Authorized Person (SAP) / Authorized Person (AP) / Competent Person (CP).

4.1 Authorization and nomination process

4.1.1 Competent Person (CP):

4.1.1.1 Minimum Requirement:

- CP is **NTPC/Outsourced Personnel** generally from Maintenance discipline. However, Operation employees working in general shift also become CP to carry out the activities assigned to Operation dept.
- ITI+ 5-year experience (post training) or Diploma+ 1-year experience (post training) or graduate engineer (post training)

4.1.1.2 Responsibilities:

- CP has to receive, transfer and clear specified safety documents such as PTW, PTW (ROMP), Limited Work Certificate (LWC), Hot work PTW etc. as per Electrical and Mechanical Safety Rules.

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- CP has to be a part of work group looking after the job.
- CP has to ensure safe conditions before start of the job.
- CP has to return the permit after he / she has inspected the job site and ensure that job is completed in all aspects and all safety requirements are complied with.
- CP's responsibility starts when he/she receives the permit and ends after permit is returned.
- For other details, refer Section-G of E&M Safety Rule Handbook- "Responsibility of Persons".
- Job safety Analysis is to be carried out by CP and submit the same to SAP. If SAP suggests any additional requirements to achieve safety from the system, those requirements are to be fulfilled.

4.1.1.3 Selection Process and Criteria

a) Minimum two-week training on

- Area specific power plant scheme tracing
- Work area specific training as required
- PTW system training.

Provision of training has been kept to familiarize new joiners to the departments, either from other departments or other power stations.

- Test is to be conducted online/offline. Nominated person has to appear in online test/offline test . PMI/RLI will develop the Body of Knowledge (BoK), Question Bank specific to work area of CP and implement the certification system.
- Those who could not qualify the test can appear as many times in next six months.
- Above process is to be repeated in case the person is transferred from main plant to Fuel Management or vice versa or in case he or she is a new entrant in maintenance function in present place of posting. In such cases, irrespective of validity of currently held certificate, a new certificate is to be issued.

After undergoing above process, station and function specific certificate of "Competent Person" will be issued by Head of the project or designated

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officer by the Head of the project to the person by name with the validity of 3 years.

4.1.2 Authorized Person (AP):

4.1.2.1 Minimum Requirement:

- *AP shall be either a Graduate engineer (post training), chemistry graduate (post training) or Diploma in engineering in relevant function (Operation/ Electrical Maintenance) (post training), or ITI (post training). All of them should have one-year experience in the relevant work.*
- AP is primarily an employee (either permanent or outsourced) working in operation of main plant, Coal handling plant, ash handling plant or water treatment plant, MGR, other offsite areas wherever applicable.
- Additionally, identified Electrical Maintenance personnel may be nominated as AP only for specific tasks¹. It is to be ensured that AP from electrical maintenance should be an employee of NTPC Ltd.,

4.1.2.2 Responsibilities:

- AP is responsible for carrying out necessary electrical and mechanical isolations for issue of safety documents in line with the discussions with Senior Authorized Person (SAP)/ Selected Person/ Control Person² as per requirement.
- “Sanctions for Test” (SFT) is to be received / cancelled by AP of Electrical Maintenance.
- Desk Engineer (Operations department) shall not be the AP in running units. A desk engineer may be approved as AP of his domain by HOP. However, the employee cannot act as AP and Desk engineer at a time.
- For other details, refer Section-G of E&M Safety Rule Handbook- “Responsibility of Persons”.

4.1.2.3 Selection process and Criteria

- a. Minimum one year in relevant job for Operation personnel. AP from Electrical Maintenance domain must have completed one year in relevant

Notes

¹To receive and clear “Sanctions for Test” in case of Electrical Apparatus, AP has to be Electrical Maintenance personnel and he/she need to be nominated by HoD-EMD. This is applicable for High Voltage Apparatus (more than 440 V only).

² Selected/Control persons, as defined in E&M Safety Rules Handbook

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job.

- b. Minimum four-week training on
 - Area specific Power plant scheme/Training,
 - Work area specific training as required
 - PTW system training.

Provision of training has been kept to familiarize new joiners to the departments, either from other departments or other power stations.

- c. Test is to be conducted for ensuring the eligibility of the person.
Nominated person has to appear in online test/offline test . The test for AP has to be area specific and should be certified for certain areas only.
- d. Those who could not qualify the test shall have to re-appear for the test and they can re appear as many times as they want in next six months .
- e. In case of outsourced persons, the person should be certified before employing him in the job.
- f. Above process is to be repeated in case the person is transferred from main plant to FM or vice versa or in case he or she is a new entrant in Operation/ Electrical Maintenance function in present place of posting. In such cases, irrespective of the validity of currently held certificate, a new certificate is to be issued.
- g. AP authorization should be area specific. For example, AP for main plant stage I units, AP for main plant stage II, AP for CHP, AP for AHP etc.,
- h. AP especially designated for high voltage level 33 KV and above should be Executive with 3 years of experience from operation/electrical Maintenance. (should be of Electrical Discipline)

Outsourced agency must submit a “Certificate of Competency” or “Electrical Work Permit”, issued by the Appropriate Government for all the persons employed as electrical technicians.

After undergoing above process, station specific certificate of “Authorized Person” will be issued by Head of the project or designated officer by the Head of the project to the person by name with the validity of three years.

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4.1.2.4 Procedure to be adopted by AP for Isolation and normalizations

- AP is responsible for all electrical and mechanical isolations and normalizations of a system for which AP is achieving safety from the system.
- AP should adhere to all the safety rules SR1 to SR11 mentioned in Section E of NTPC mechanical and electrical safety rules.
- AP should use PPEs mandatory for electrical isolation / normalization (Electrical arc suit, face shield & hand gloves)
- AP should use LOTO kits for all isolations carried out by him.

4.1.3 Senior Authorized Person (SAP)

4.1.3.1 Minimum Requirement:

- SAP should be an NTPC executive with Degree in Engineering / Post Graduate degree in Chemistry.
- SAP is an Operation (Main plant / CHP / MGR / AHP/ BoP/ Chem) personnel.
- He / she has operational experience of minimum three years.
- For issuing PTW in systems on and above 33 KV ASCE/SCE with 5 years of operation experience is only eligible to issue PTW.

4.1.3.2 Responsibilities:

- SAP has to prepare, issue, suspend, transfer and cancel the safety documents such as PTW, PTW(ROMP), LWC and SFT.
- SAP can undertake the role of AP if required, however, he or she cannot exercise both the roles (AP and SAP) in the same safety document.
- For other details refer Section-G of E&M Safety Rule Handbook- "Responsibility of Persons".

4.1.3.3 Procedure to be adopted by SAP for issue of safety documents

- SAP has to discuss with CP and selected persons about various safety aspects involved in the work to be carried out and understand the requirements to be fulfilled for achieving safety from the system.
- SAP communicates with CP regarding the requirements for achieving

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safety from the system. If any requirements are to be fulfilled in addition to requirements of CP, those requirements also to be included in safety documents.

- SAP may discuss with selected persons regarding the Works to be carried out and requirements for achieving the safety from the system
- SAP gives written communication to AP regarding the isolations and safety precautions to be applied
- SAP accepts communication from CP and AP in writing only.
- SAP maintains the safety Keys and cross permits to be issued on the system
- If one SAP prepares safety document and another SAP can issue the permit. In this case achieving the safety from the system will be on both SAPs.
- SAP maintains a register of Safety documents issued and handover them to the next SAP with full details mentioned in the register and safety documents.
- IF PTW (ROMP) is issued by SAP, no other safety document is to be issued by anybody on the same system
- IF SFT is issued then preparation and issue are to be done by one SAP only.

4.1.3.4 Selection process and Criteria

- Minimum three years in Operation of the equipment in the area where he is going to be SAP. Commissioning experience in the same area shall also be considered as relevant experience.
- Minimum Two-week training on
 - Area specific Power plant scheme/Training,
 - Area specific training
 - PTW system training.
- Provision of training has been kept to familiarize new joiners to the departments, either from other departments or other power stations. Duration of training can be extended to maximum one month at the request of prospective SAP if he / she desires so.
- Test is to be conducted online/offline. Nominated person has to appear in online test/offline test.

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- e. Those who could not qualify the test shall have to re-appear for the test within next six months.
- f. Above process is to be repeated in case the person is transferred from main plant to FM or vice versa or in case he or she is a new entrant in Operation function in present place of posting. In such cases, irrespective of the validity of currently held certificate, a new certificate is to be issued.

After undergoing above process, station specific certificate of “Senior Authorized Person” will be issued by Head of the project or designated officer by the Head of the project to the person by name with the validity of three years. SAP should also to be specified area wise.

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Key Notes

- a. All employees fulfilling the eligibility criteria have to be certified as either CP/AP/SAP as per requirement.
- b. One person cannot be both CP and AP.
- c. PMI training scheme for ITI/Diploma/graduate engineer to take care of such needs so that the participants are equipped with competencies required to be demonstrated in certification examinations.
- d. PMI/RLI is responsible for design, development and delivery of Two-week training module for CP/AP/SAP for existing employees.
- e. For the online test / offline test to be conducted for certifying a Person as CP/AP/SAP
 - PMI/RLI will develop the Body of Knowledge (BoK), Question Bank and implement the certification system. In the online system, certain percentage of questions from each section of BoK shall be randomly picked up to be posed to the aspirants.
 - PMI/RLI is responsible for preparation, periodic review, update and deploying randomized question bank as well as conducting the test with the help of IT group.
 - Question bank has to take into account general technical competency and plant specific technical requirements.
 - Site EDC/RLI to conduct the test and communicate the marks to certifying authority.
 - Any employee should be able to take the certification test on

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demand as and when required.

- PMI should prepare the SOP for certifying the CP/AP/SAP with proper approval from the relevant authority.
- f. Validity of CP/AP/SAP certificates shall extend up to end of the month of validity end date.
- g. A comprehensive, updated and current list/record of all certified Competent/ Authorized/ Senior Authorized persons must be maintained and made available by PMI and should be linked to GRC framework in ERP system.

43 General guidelines for Permit system

4.3.1 There are four kinds of permits

4.3.1.1 PTW

4.3.1.2 PTW(ROMP)

4.3.1.3 LWC

4.3.1.4 SFT

4.3.2 Any one of these types of permits are to be issued depending on type of work to be done. The kind of permit is to be issued is in the authority of SAP.

4.3.3 Any one of these permits may be issued along with the above permit

4.3.3.1 Work at height

4.3.3.2 Work inside confined spaces

4.3.3.3 Hot work permit

4.3.3.4 Excavation permit

The above permits are to countersigned by the respective identified authorized persons by HOP after checking the necessary requirements for achieving the safety from the system. Any other permit as applicable based on the system requirements like under water working etc., or coming up in future shall also be issued in the same manner.

4.3.4 The permit area is defined in SAP as train of the system. SAP are to be defined for a particular train. If a particular SAP can be SAP for two or more trains they are to be clearly specified by the head of station in his authorization certificate.

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- 4.3.5 If some of the isolations and normalizations are to be done in some other areas, then SAP of that particular area has to send written request of isolation and normalization to AP of the area where isolation or normalization to be done. Then AP on receipt of such request will process the request and achieve the safety from the system in his area and send all the requisite keys and fuses to SAP who has requested and the same is mentioned in the written request as well.
- 4.3.6 SAPs should be decided based on the control centers used as per the plant requirements like Main plant units SAP should be different from Ash handling system SAP if they are operated from different control rooms. Similarly, SAP for CHP should be different from main plant area if control rooms are different.
- 4.3.7 Safety from the system is achieved by SAP in consultation with competent person, Authorized persons and selected persons
- 4.3.8 If any confusion is there regarding scope of work and any other aspects, SAP should consult the area experts and selected persons and issue the safety documents once safety from the system is achieved.
- 4.3.9 Limited work certificate permits are the most critical permits for the system. These safety documents are to be issued after SAP understands the complete safety and operational aspects of the system. These permits are to be operated with utmost care as most of the LWCs require working on the live equipment's.
- 4.3.10 Once PTW (ROMP) is issued no other permits are to be issued on that equipment.
- 4.3.11 PTW and LWC on the same equipment should not be issued without proper assessment of the scope of work and safety aspects.
- 4.3.12 Job safety analysis is to be carried out by competent person in consultation with selected persons and this JSA has to be submitted to SAP. SAP after studying the JSA may accept the JSA as it is or may request CP to include any other safety aspect. CP should change the JSA accordingly and PTW to be issued once JSA is accepted by CP and SAP.
- 4.3.13 For every Control Centre, a PTW register is to be maintained with all details regarding the PTWs, pending PTWs, Cross Permits.
- 4.3.14 SAP should update this register as and when PTW is issued and cancelled or transferred.

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5.0 EXCEPTIONS

In case of any deviation with regard to deployment of CP/AP/SAP, necessitated due to local conditions, approval can be accorded by HOP with proper justification and under intimation to ED (OS) and ED (Safety).

6.0 RESPONSIBILITY

Responsibility of nominations for SAPs, APs & CPs is with HOD of respective departments. Individual HODs are responsible for getting the certification of CP/AP/SAP to persons working under them for the PTW system.

7.0 RECORD KEEPING

HODs are responsible for keeping the record of SAPs, APs & CPs of their sections. MTP is responsible for keeping consolidated list of of SAPs, APs & CPs of total O&M sections.

8.0 REVIEW

The Head of the Project will be responsible for reviewing this document on a Two yearly basis

Annexure A – List of Abbreviations

NTPC	LMI	SIMHADRI
TITLE	LMI ON CERTIFICATION OF SAPs, APs & CPs FOR PTW SYSTEM	DATE: OCT 2022
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AP	Authorized Person
AHP	Ash Handling Plant
ASCE	Assistant Shift Charge Engineer
BoK	Body of Knowledge
BoP	Balance of Plant
CP	Competent Person
CHP	Coal Handling Plant
DT	Diploma Trainee
EET	Engineering Executive Trainee
EDC	Employee Development Centre
EMD	Electrical Maintenance Department
ERP	Enterprise Resource Planning
E&M	Electrical & Maintenance
FM	Fuel Management
GRC	Governance, Risk and Compliance
HOD	Head of Department
HOP	Head of Project
LWC	Limited Work Certificate
MGR	Merry Go Round
OGN	Operation Guidance Note
O&M	Operation & Maintenance
PMI	Power Management Institute
PMS	Performance Management System
PTW	Permit to Work
RLI	Regional Learning Institute
ROMP	Restoration of Motive Power
SAP	Senior Authorized Person
SCE	Shift Charge Engineer
SFT	Sanction for Test

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Annexure B

Snapshot of Roles of CP/AP/SAP

Role	Function	Qualification/Exp	Pre-requisites	Selection Criteria
Competent Person (CP)	• NTPC/outsourced personnel. • Maintenance Personnel. • Operation personnel working in general shift	ITI with 5-year experience (post training) / Diploma with 1-year experience (post training)/ Graduate Engineer (post training)	All employees should have completed training	• Online test/ Offline test
Authorized Person (AP)	• NTPC/ Out sourced Personnel. • Operation/ Identified Electrical Maint. Personnel.	Diploma with 1-year experience (post training)/ Graduate Engineer (post training)/ Chemistry graduate (post training)/ITI with 1 year experience in relevant function(post training)	Outsourced manpower should undergo four weeks training before going for test. NTPC employees should undergo compulsory training before going for test.	• Online test/ Offline test
Senior Authorized Person (SAP)	• NTPC Personnel. • Operation Personnel.	• Executive (Graduate Engineer / post Graduate in Chemistry) with 3-year experience in operation of the relevant area (post training)	• All employees should have completed training	• Online test/ Offline test

Notes: 1. Desk Engineer of a running unit cannot be a signatory to any safety document.

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Annexure C

Definitions

Safety from the system: that condition Which safeguards persons working on our testing plant and/or apparatus from the dangers which are inherent in the system.

Competent person

A person who has sufficient technical knowledge and/or experience to enable him who avoid danger and who may receive, transfer and clear specified safety document then nominated by the Head of O&M.

Authorized person:

A competent Person who has been trained and then authorized by the location Head of O&M in writing to carry out specified duties.

Senior Authorized person

An Authorized person nominated by the Head of O&M and trained to carry out duties specified in writing. Including the preparation, issue and cancellation of specified documents.

Control person:

A person who has been nominated by the Head of O&M to be responsible for controlling and coordinating safety activities necessary to achieve safety from the system.

Selected person:

A person qualified by technical knowledge and experience and selected by the Head of O&M to carry out the tests and examinations and make recommendations regarding additional special Precautions to be taken to safeguard persons.

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ANNEXURE D

Qualifications of the persons engaged as AP for carrying out electrical Isolations:

1. The persons used for carrying out the electrical switching operations and isolations may be NTPC employees or manpower supplied through Outsourced agencies with minimum qualifications as mentioned in section 4.1.2
2. General qualifications applicable to all
 - a. The person employed shall be either Diploma in engineering as approved by AICTE or ITI two year course in relevant trade as approved by state governments.
3. In case of NTPC employees:
 - a. He Should be an NTPC employee recruited specifically for electrical technician and should have minimum one-year experience post training.
 - b. He should be nominated for carrying out the electrical isolations by head of O&M in consultation with head of operation.
 - c. His skills are to be monitored periodically at least once in six months and training should be imparted every year.
 - d. The authorization is to be limited to the area where his experience permits. Ex. He should be employed in 6.6 KV and 415V unit and station switchgears only if his experience is only in these switchgears.
4. In case of personnel employed through outsourced agencies
 - a. He should undergo training module of four weeks in RLI. the typical syllabus and course content are issued to RLI/PMI. RLI/training Centre should use this course content and may enhance the course content as per site requirements.
 - b. After this module the person will undergo a test conducted by a committee of three executives (one AGM rank as the head and two not less than E6 rank (one from operation and other from Electrical maintenance)). The committee will test the practical skill and required theoretical knowledge and submit their recommendation to Head of O&M.
 - c. Outsourced agency must submit a “Certificate of Competency” or “Electrical Work Permit”, issued by the Appropriate Government for all the persons employed as electrical technicians
 - d. After successful completion of training, GM(O&M) permits the outsourced agency to deploy that particular person in the job of carrying out isolations and normalizations for specialized training for on job in parallel with existing employees for another eight weeks and will be under strict monitoring of AP. Then they can be used for electrical isolations and normalizations.